Data Science

CSE558

Overview

- z-test for population mean and population proportion

- t-test

Overview

- Hypothesis Testing

Question

1. A manager at a cafe notices that 15% of customers orders more food while having coffee/tea. In an effort to increase sales, the restaurant begins offering free refills with every coffee/tea order for a trial period of two weeks. During this time 22% of customers ordered more food while having coffee/tea. To test if the promotion was successful in increasing cafe's sale, what will be the hypotheses that the manager will write?
2. Let your friend claims on an average she can correctly guess the suit of a randomly picked card with more than 0.25 probability. In your experiment, if she correctly guessed suits with 0.28 probability then to reject your null hypothesis how many trials you have to see?

χ^2-test for Population Variance

- Univariable test, when sample is assumed from a normal distribution.
- Investigate sample variance vs population variance
- Square of normal distribution is χ^2 distribution.

From a population of assumed variance σ_0^2, a random sample of size n is taken.
H_0: $\sigma^2 = \sigma_0^2$; H_1: $\sigma^2 \neq \sigma_0^2$;
Level of significance: $\propto = 0.05$

Let sample variance is s^2. Then the test statistics. $\chi^2 = \dfrac{(n-1)s^2}{\sigma_0^2}$

Degrees of freedom = n-1

Example

In the last month, the standard deviation of the train delays at your station was reported to be 2 hours. In this month, you took a random sample of 17 trains and noticed a standard deviation of $\sqrt{7}$ hours. With 5% level of significance do you think the reliability of Indian railway has changed or still the same.

Example

From a population of assumed variance σ_0^2 = 4, a random sample of size n =17 is taken.

H_0: $\sigma^2 = \sigma_0^2$; H_1: $\sigma^2 \neq \sigma_0^2$; Level of significance: $\propto$ = 0.05

Let sample variance is s^2 = 7. Then the test statistics χ^2 = 28.

Two-sided One-sided	0.20 0.10		0.10 0.05		0.05 0.025		0.01 0.005	
ν	a	b	a	b	a	b	a	b
1	0.016	2.71	39.10^{-4}	3.84	98.10^{-5}	5.02	16.10^{-5}	6.63
2	0.21	4.61	0.10	5.99	0.05	7.38	0.02	9.21
3	0.58	6.25	0.35	7.81	0.22	9.35	0.11	11.34
4	1.06	7.78	0.71	9.49	0.48	11.14	0.30	13.28
5	1.61	9.24	1.15	11.07	0.83	12.83	0.55	15.09
6	2.20	10.64	1.64	12.59	1.24	14.45	0.87	16.81
7	2.83	12.02	2.17	14.07	1.69	16.01	1.24	18.48
8	3.49	13.36	2.73	15.51	2.18	17.53	1.65	20.09
9	4.17	14.68	3.33	16.92	2.70	19.02	2.09	21.67
10	4.87	15.99	3.94	18.31	3.25	20.48	2.56	23.21
11	5.58	17.28	4.57	19.68	3.82	21.92	3.05	24.73
12	6.30	18.55	5.23	21.03	4.40	23.34	3.57	26.22
13	7.04	19.81	5.89	22.36	5.01	24.74	4.11	27.69
14	7.79	21.06	6.57	23.68	5.63	26.12	4.66	29.14
15	8.55	22.31	7.26	25.00	6.26	27.49	5.23	30.58
16	9.31	23.54	7.96	26.30	6.91	28.85	5.81	32.00

Degrees of freedom = 16

Conclusion: Do not reject *H_0*.

Example

A bank with a multiple customer support window finds that the standard deviation for normally distributed waiting times for customers on Saturday afternoon is 9 minutes. The bank experiments with a single window and finds that for a random sample of 31 customers, the waiting times for customers have a standard deviation of 6 minutes.

With 5% level of significance, test claims that one window lowers variation among waiting times.

Degrees of freedom (*df*)	Significance level (α)							
	.99	.975	.95	.9	.1	.05	.025	.01
1	-------	0.001	0.004	0.016	2.706	3.841	5.024	6.635
2	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210
3	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345
4	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277
5	0.554	0.831	1.145	1.610	9.236	11.070	12.833	15.086
6	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812
7	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475
8	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090
9	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666
10	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209
11	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725
12	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217
13	4.107	5.009	5.892	7.042	19.812	22.362	24.736	27.688
14	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141
15	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578
16	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000
17	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409
18	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805
19	7.633	8.907	10.117	11.651	27.204	30.144	32.852	36.191
20	8.260	9.591	10.851	12.443	28.412	31.410	34.170	37.566
21	8.897	10.283	11.591	13.240	29.615	32.671	35.479	38.932
22	9.542	10.982	12.338	14.041	30.813	33.924	36.781	40.289
23	10.196	11.689	13.091	14.848	32.007	35.172	38.076	41.638
24	10.856	12.401	13.848	15.659	33.196	36.415	39.364	42.980
25	11.524	13.120	14.611	16.473	34.382	37.652	40.646	44.314
26	12.198	13.844	15.379	17.292	35.563	38.885	41.923	45.642
27	12.879	14.573	16.151	18.114	36.741	40.113	43.195	46.963
28	13.565	15.308	16.928	18.939	37.916	41.337	44.461	48.278
29	14.256	16.047	17.708	19.768	39.087	42.557	45.722	49.588
30	14.953	16.791	18.493	20.599	40.256	43.773	46.979	50.892
40	22.164	24.433	26.509	29.051	51.805	55.758	59.342	63.691
50	29.707	32.357	34.764	37.689	63.167	67.505	71.420	76.154
60	37.485	40.482	43.188	46.459	74.397	79.082	83.298	88.379
70	45.442	48.758	51.739	55.329	85.527	90.531	95.023	100.425
80	53.540	57.153	60.391	64.278	96.578	101.879	106.629	112.329
100	61.754	65.647	69.126	73.291	107.565	113.145	118.136	124.116
1000	70.065	74.222	77.929	82.358	118.498	124.342	129.561	135.807

z-test for Two Population Mean

- Tests significance of the difference between the means of two populations.
- Standard deviations of both populations are known.
- Useful if populations are normally distributed (Large Samples).

From two population of assumed mean μ_1 & μ_2 and known standard deviations σ_1 & σ_2, two random sample of size n_1 & n_2 are taken from both populations.

H_0: $\mu_1 = \mu_2$; H_1: $\mu_1 \neq \mu_2$;

Level of significance: $\propto = 0.05$

Let the sample means be $\bar{x}_1, \bar{x}_2$, then the test statistics $Z = \dfrac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\left(\dfrac{\sigma_1^2}{n_1} + \dfrac{\sigma_2^2}{n_2}\right)^{\frac{1}{2}}}$

Limitations: Accurate when distribution normal, else the test approximates.

Example

On the following, with a 5% level of significance test if the height of US and Swedish males are different.

	Height (Inches)			Height (Inches)			Height (Inches)	
ID	US	Swedish	ID	US	Swedish	ID	US	Swedish
1	69.12	74.56	11	69.18	68.00	21	72.12	72.00
2	66.88	71.89	12	66.18	72.00	22	66.88	70.00
3	74.82	73.00	13	64.94	73.56	23	73.82	69.22
4	67.00	67.78	14	71.76	72.56	24	74.00	74.44
5	69.12	72.22	15	70.12	75.00	25	71.18	68.00
6	65.00	68.00	16	71.00	68.33	26	67.88	73.89
7	71.00	73.56	17	71.88	71.67	27	65.94	70.00
8	66.76	75.00	18	65.24	72.44	28	68.88	70.44
9	72.12	68.22	19	70.06	75.00	29	68.00	70.22
10	72.94	69.00	20	71.94	71.89	30	75.12	73.33

Calculation	US male	Swedish Male
Sample mean $\bar{x}$,	69.70	71.51
Population Standard Deviation	3.12	2.44
Sample Size, n	30	30

z-test for Two Proportions

- To test if the proportions in two populations are equal.
- Useful when the sample size is large and standard deviations are known
- Accurate when distribution normal, else the test approximates.

From two population of assumed success rates μ_1 & μ_2, two random sample of sizes n_1 & n_2 are sampled respectively.
$H_0: \mu_1 = \mu_2$; $H_1: \mu_1 \neq \mu_2$;
Level of significance: $\propto = 0.05$
Let the success rates in each samples be p_1 and p_2. Then the test statistics is

$$Z = \frac{(p_1 - p_2)}{\left\{P(1-P)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)\right\}^{\frac{1}{2}}}$$

$$P = \frac{p_1 n_1 + p_2 n_2}{n_1 + n_2}$$

Example

Vaccine Testing: Out of 900 people in experimental group 899 of them does not contract virus after vaccination and out 1600 people in the control group 1576 people does not contract the virus. With a 5% level of significance, do you think your vaccines reduces virus contraction rate?

t-test for Two Population Mean

- Investigate difference between two population means
- Standard deviation of populations are unknown but known to be equal.

From two population of assumed mean μ_1 & μ_2, two random sample of sizes n_1 & n_2 are sampled respectively.
H_0: $\mu_1 = \mu_2$; H_1: $\mu_1 \neq \mu_2$;
Level of significance: $\propto = 0.05$

Let sample means be $\bar{x}_1, \bar{x}_2$ and $s^2 = [(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2]/(n_1 + n_2 - 2)$, where s_1^2 and s_2^2 are the variance of the sample. Then test statistics $t = \dfrac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{s\left(\dfrac{1}{n_1} + \dfrac{1}{n_2}\right)^{\frac{1}{2}}}$

Degrees of freedom = $n_1 + n_2 - 2$.

Example

Consider the following body fat % data by genders. With 5% level of significance test if body fat % in men is significantly different from that in men.

Men	13.3	-6	20	8	14	19	18
	25	16	24	15	1	105	
Women	22	16	21.7	21	30	26	
	12	23.2	28	23			

ν	Level of significance α 0.10	0.05	0.025	0.01	0.005
1	3.078	6.314	12.706	31.821	63.657
2	1.886	2.920	4.303	6.965	9.925
3	1.638	2.353	3.182	4.541	5.841
4	1.533	2.132	2.776	3.747	4.604
5	1.476	2.015	2.571	3.365	4.032
6	1.440	1.943	2.447	3.143	3.707
7	1.415	1.895	2.365	2.998	3.499
8	1.397	1.860	2.306	2.896	3.355
9	1.383	1.833	2.262	2.821	3.250
10	1.372	1.812	2.228	2.764	3.169
11	1.363	1.796	2.201	2.718	3.106
12	1.356	1.782	2.179	2.681	3.055
13	1.350	1.771	2.160	2.650	3.012
14	1.345	1.761	2.145	2.624	2.977
15	1.341	1.753	2.131	2.602	2.947
16	1.337	1.746	2.120	2.583	2.921
17	1.333	1.740	2.110	2.567	2.898
18	1.330	1.734	2.101	2.552	2.878
19	1.328	1.729	2.093	2.539	2.861
20	1.325	1.725	2.086	2.528	2.845
21	1.323	1.721	2.080	2.518	2.831
22	1.321	1.717	2.074	2.508	2.819
23	1.319	1.714	2.069	2.500	2.807

t-test for Two Population Mean (Paired)

- Investigate difference between the means of a pair of population.
- Standard deviation of populations are unknown.

From two population of assumed mean μ_1 & μ_2, pair of n random sample of size *n*.
H_0: $\mu_1 = \mu_2$; H_1: $\mu_1 \neq \mu_2$;
Level of significance: $\propto = 0.05$

Let sample means be $\bar{x}_1, \bar{x}_2$ and $s^2 = \sum_{i=1}^{n} \frac{(d_i - \bar{d})^2}{n-1}$

Then, the test statistics $t = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{s/n^{\frac{1}{2}}}$. Degrees of freedom = n-1